

The Microenvironment of Wound Healing

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Abstract: This review summarizes experiments performed by us and others, examining the importance of the microenvironment to wound healing. The development of a sealed polyurethane wound chamber has allowed us to perform studies evaluating the effects of growth factors, transplanted cells, and other bioactive substances on wound healing. Studies have compared wet, moist, and dry healing, with the conclusion that a wet, incubator-like microenvironment provides the fastest healing with fewest aberrations and least scar formation. The wet environment is also paramount for the survival and proliferation of transplanted cells or tissue, which has been shown in studies of porcine and human wounds. Moreover, high concentrations of antibiotics and other agents can be introduced in the wound chamber, thereby effectively fighting infection, while maintaining safe systemic concentrations. These findings have been used in clinical settings to treat wounds of different types. A titanium chamber has been developed to create an *in vivo* incubator, which will serve as a regenerative platform for *in vivo* tissue engineering.

Key Words: Microenvironment, wound healing, skin transplantation, tissue engineering

(*J Craniofac Surg* 2013;24: 12–16)

Human skin has a large surface area-to-volume ratio. This makes it hard to precisely control the skin wound microenvironment. However, through the construction of wound chambers that can be attached to the skin with adhesives, a sealed, controlled environment can be created. This procedure creates an *in vivo* incubator into which cells, tissue constructs, bioactive molecules, and specialized culture media can be introduced.

There are several good reasons to learn to control the skin microenvironment. One is that experimental and clinical treatment of wounds can be standardized. Another reason is that the microenvironment can be monitored by sampling the liquid adjacent to the wound, because the wound interior will be in constant exchange with the wound fluid. Third, the controlled environment allows the delivery of therapeutic agents such as antibiotics in high concentrations through the liquid. The results from such treatment show that wounds heal faster and with less scarring than in either a moist or dry environment.

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Received August 20, 2012.

Accepted for publication August 25, 2012.

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Dr. Eriksson is a member of a limited liability company that receives royalty payments from wound chamber sales. The other authors report no conflicts of interest.

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ISSN: 1049-2275

DOI: 10.1097/SCS.0b013e31827104fb

Our group has studied the wound microenvironment and its influence on healing for the past few decades. We have been able to validate the methodology discussed below and expand its potential into the areas of delivery of growth media, therapeutic agents, proteins, and genes, as well as cells and skin micrografts. This report summarizes some of the results from these studies and discusses the detailed results of others.

Definition of the Wound Microenvironment

The wound microenvironment is defined as the environment exterior to the wound and in direct contact with its surface. It is defined as *dry* when there is no barrier to contain the water and solids in the wound. It is described as *moist* when a moisture-containing controlled hydration dressing is used to cover the surface of the wound. Finally, it is described as *wet* when covered with an impermeable membrane that is sealed to the periphery of the wound with adhesives.

Wound Chamber Designs

The microenvironment of the wound has been studied in a number of experiments, most of them involving porcine wounds, but some in treating the wounds of human patients. In early experiments, a transparent vinyl wound chamber was attached to the perimeter of the wound with an adhesive. Later, polyurethane chambers of comparable design were used. In porcine experiments, the chambers usually stay intact without any leakage for 4 to 7 days after placement. In humans, it is easier to keep the chambers adherent for a longer time without any disruption or leakage, usually 6 to 7 days. Current strategies to study and modulate the wound microenvironment include the development of an enclosed titanium chamber model in animals (Fig. 1).¹

The titanium chamber is implanted by making an incision through the skin and is stabilized with sutures in the subcutaneous muscle layer. The skin incision is closed around the cylindrical part of the chamber using a purse-string suture. Cells, tissue constructs, bioactive molecules, and specialized culture media can be introduced into the chamber (Fig. 2). The chamber/*in vivo* incubator can be used to study the process of wound healing in a highly standardized fashion, with continuous sampling of the microenvironment via a removable lid. The chamber enables studies of tissue integration following transplantation (Fig. 2). Current attempts to generate tissue constructs *in vitro* have yet to overcome the obstacle of neovascularization. The limited diffusion distance of oxygen and nutrients in tissue restricts the size of engineered constructs to a few millimeters. By relying on the ingrowth of blood vessels from the wound bed, a vascularized tissue construct can be engineered in the chamber.

Wet Wound Healing

In 1861, Hebra² published a report on the treatment of burns in a "continuous bath." He treated patients with major burns in bathtubs for months and even years. Hebra reported that treatment in water reduced the patients' pain, attenuated their weight loss, and literally kept them alive; he also reported that the patients died when

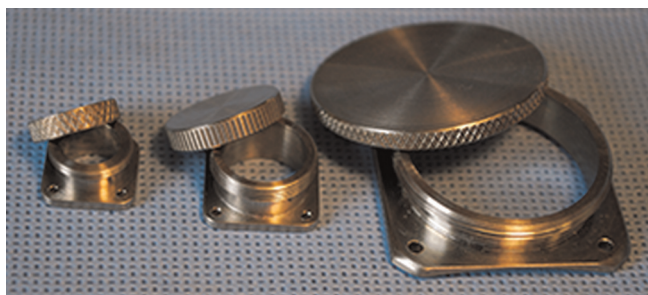


FIGURE 1. Custom-made titanium chambers. Titanium chambers for implantation in mice (left), rat (center), and pig (right). The design incorporating a screw-on lid enables continuous monitoring of the wound microenvironment and delivery of bioactive substances.

the treatment in water was discontinued. This work predated the development of skin grafting. During World War II, John Bunyan, a medical officer and lieutenant in the British Navy, treated many wounded soldiers with procedures based on an irrigation method using bleach.³ When he used the irrigation fluid that contained bleach, he noticed less necrosis as well as a reduced rate of infection. Since Bunyan's report, many other irrigation systems have been described, for instance, by Svedman,⁴ Kinetic Concepts Incorporated (KCI),⁵ and Zamirowski.⁶ Different antimicrobial solutions have been used with variable success. Collectively, these efforts illustrate the difficulties of producing an effective device as a vehicle for the liquid. They also illustrate the difficulties of delivering precise amounts of antibiotics via an irrigation system. In practice, the concentrations of antibiotics are kept quite low to avoid the possibility of systemic toxicity.

In an early study, the influence of the liquid on partial thickness wounds was examined.⁷ The liquid in the chamber was usually approximately 0.2 mL of nonbuffered normal saline per square centimeter of the wound. Tissue culture concentrations of penicillin and streptomycin were used to prevent bacterial growth within the chamber. Within 48 hours of applying the wound chamber, the amount of liquid in the wound had decreased by approximately half, most likely because of the low oncotic pressure of the saline. At 48 hours, the wound fluid contained 3 to 4 g/dL of protein and various electrolytes at concentrations similar to extracellular fluid. The protein concentration in the wound chamber was found to decrease progressively as the surface epithelium of the wound developed.⁷ The protein concentration in the liquid could therefore be used as a reliable and highly reproducible noninvasive monitor of re-epithelialization.

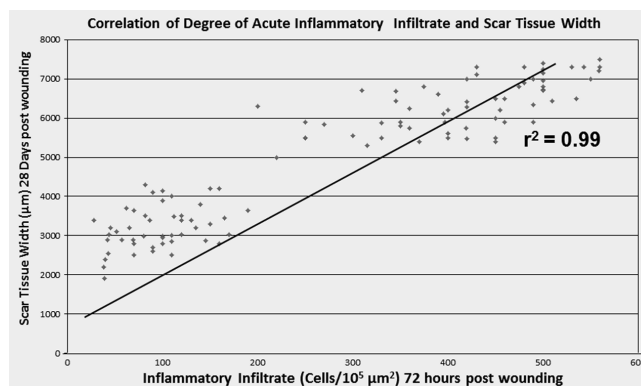


FIGURE 3. Correlation of degree of acute inflammatory infiltrate and scar tissue width. Treatment of porcine wounds with a wound chamber reduced inflammation and subsequent scar formation as compared with treatment with dry gauze.

Further experiments were also performed to study the flux of fluid across the wound surface. As mentioned earlier, when a hypo-oncotic solution such as saline was used in the chamber, the amount of chamber fluid decreased. If, on the other hand, hyperoncotic fluid such as Dextran 40 in a 10% concentration (approximately 10 times the oncotic pressure of plasma) was used in the chamber, the volume increased as predicted because of the oncotic force of the macromolecules in the fluid. From these experiments, it was concluded that the wound surface acts as a highly permeable membrane.

The Impact of the Microenvironment on Scar Formation

In a study by Reish and colleagues,⁸ healing was studied in both wet (chamber-enclosed) and dry environments. The dry environment was associated with considerably more inflammation as well as a significantly wider scar. The degree of inflammation as quantified by the number of inflammatory cells in the early wound (72 hours after wounding) correlated strongly with the width of the scar at 28 days after wounding (Fig. 3). In the wet environment, there were fewer aberrations from normal skin in terms of the formation of rete ridges. Control of the microenvironment also provides major opportunities for early intervention to minimize scarring. This is an area in which future studies will provide useful information.

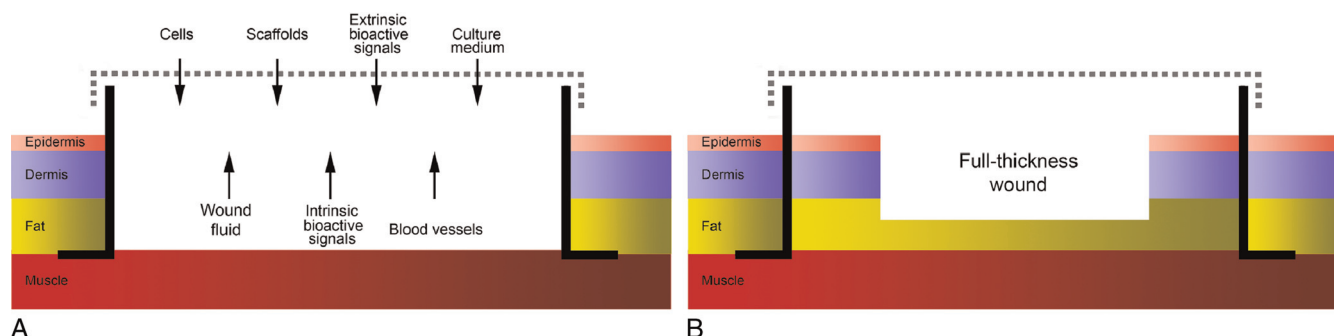


FIGURE 2. Chamber for tissue engineering and wound healing studies. A, After implantation, the wound chamber (black lines) and the screw-on lid (dotted lines) create a sealed wound microenvironment. Wound fluid and ingrowing vessels enter the chamber from the wound bed. Cells, scaffolds, bioactive signals, defined culture medium, or other substances can be introduced to the healing wound. Fusenig and collaborators first described this type of chamber made of a plastic material for use in cancer studies.^{7a} B, Eriksson designed it in titanium with a removable lid, with a skin island and a central wound inside.

The multiplication of microorganisms in the chamber, with potential invasion into the wound tissue, has traditionally been a concern when using a completely occlusive treatment with a considerable volume of liquid. The effect of hydration alone has also been a concern. Our experiments show that hydration alone without a high degree of bacterial contamination or invasion is not deleterious to the wound or the surrounding skin.⁷ The wounds will heal faster in a wet noncontaminated environment than in a moist or dry environment.^{9,10} In addition, a seemingly normal stratum corneum will form in this environment. When a small amount of antibiotics is present in the wound fluid (in our case, 100 U of penicillin and 100 µg of streptomycin per milliliter of added wound fluid in the amount of 0.2 mL/cm² daily), the quantitative cultures of the wound show either no growth or a maximum of 10² of bacteria.¹¹ If no antibiotics are added to the liquid in the wound chamber, the number of the bacteria per milliliter of wound fluid will increase to an average of 10⁵. Most of these bacteria are noninvasive *Staphylococcus albus*. The corresponding tissue biopsies will show very low bacterial counts, usually a maximum of 10³. If highly invasive bacteria are introduced into the chamber, and no antibiotics are added, this will result in an invasive infection with counts 10⁵ to 10⁶ of bacteria per gram of tissue.¹² It is quite feasible to introduce antimicrobials of different kinds into the wound chamber, and this will rapidly decrease wound fluid and tissue bacterial counts to very low levels or even zero.

Analysis of the common growth factors in the wound fluid revealed them to be present in concentrations similar to those found in plasma.¹³ A large number of experiments were done with delivery of physiological or supraphysiological concentrations of growth factors. Two of them, epidermal growth factor and platelet-derived growth factor, elicited a significant increase in healing of partial-thickness wounds.¹³ When epidermal growth factor was delivered by gene transfer, there was a similar acceleration in healing of partial-thickness wounds.¹⁴

Delivery of Antibiotics

Various molecules can be injected into the microenvironment of the wound through a port or tube in the chamber. The chamber design also allows for sampling of wound fluid to determine the rate of elimination of the antibiotic within the chamber itself. In a human study, Vranckx et al¹⁵ studied 28 wounds, most of which had failed conventional treatment (Fig. 4). The treatment algorithm consisted of debridement, application of a wound chamber, and addition of culture-guided antibiotics at high concentrations. The treatment was continued for 3 to 7 days preceding either skin grafting or flap closure. In some cases, the wound chamber treatment was continued until the wound was healed. Of the 28 wounds treated in this way, 25 healed. In the other 3 cases, it was repeated for a second session.

It was found that antibiotics effective against common bacteria could easily be delivered in concentrations up to and exceeding



FIGURE 4. Treatment of infected lower-extremity wound with open fracture using wound chamber and antibiotics. A, After debridement, the wound is prepared for immediate closure. B, Early wound chamber prototype. C, Wound after skin grafting following 2 weeks of treatment with the wound chamber. (Modified and reprinted with permission from Lippincott Williams & Wilkins.)

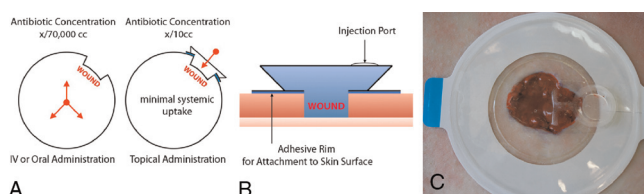


FIGURE 5. Immediate wound enclosure using a wound chamber that delivers ultrahigh concentrations of antibiotics. A, In the average person, a given dose of antibiotics is diluted in 70,000 mL (volume of a 100-kg man) when delivered intravenously or orally. When delivered directly or topically, this same dose is diluted only in the wound fluid exudate and can therefore provide a concentration that is 7000 times higher and in direct contact with bacterial contamination. B, Schematic of a wound chamber including injection port and adhesive rim for attachment to the skin surface. C, Wound chamber, composed of polyurethane with an adhesive flexible ring and access port to allow delivery of drugs (eg, antibiotics, antifungals, and analgesics).

1000 times minimum inhibitory concentration (MIC) for common bacteria and antibiotics (Fig. 5). The general principle was that no more than 1 standard intravenous dose for that particular patient was delivered at one time. There were no adverse reactions other than slight pain when gentamicin was introduced into the chamber. The pain could easily be eliminated with a small dose of lidocaine in the wound fluid. No other antibiotic caused pain.

With the concentrations of gentamicin, clindamycin, cephalixin, and vancomycin used (which varied between 100 and 1000 times MIC), the rate of healing did not seem to be impaired. One patient with a deep abdominal fistula was treated for 72 days until the fistula was closed. After approximately 3 weeks, there was an overgrowth of *Candida albicans* in the wound chamber, which was easily eliminated with a small dose of amphotericin B. Some of the patients had severe pain in the area of the wound that could be

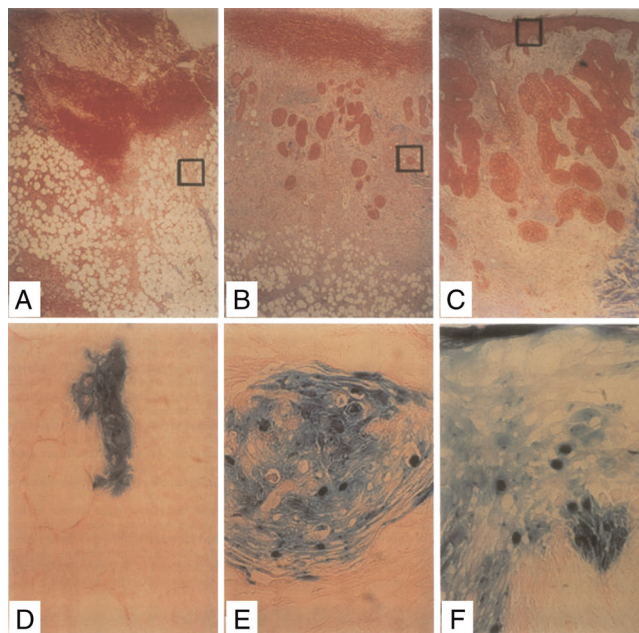


FIGURE 6. Reconstitution of new epithelium of full-thickness wounds. A–C, Cross sections stained with Masson trichrome. D and E, Areas in squares showing X-Gal staining at higher magnification, demonstrating LacZ expression. As can be seen in the images, the cells migrated upward and are present in all layers of the epithelium at day 8. (Reprinted with permission from *Proceedings of the National Academy of Sciences*, Stanford University's Highwire Press.)

alleviated with a small dose of lidocaine in the chamber. It should be noted that treatment with antibiotics indicated for intravenous use would be “off-label use” if used in the wound chamber.

Transplantation of Cells and Micrografts

Vogt et al¹⁶ studied the transplantation of cells into the wound chamber microenvironment. It was found that when a single-cell suspension of keratinocytes was injected into the chamber, these cells would survive, proliferate, migrate to the surface of the wound, and form a new epidermis with a basement membrane and basal layer, and they seemingly undergo the normal process of differentiation (Fig. 6). This was confirmed by gene markers (LacZ). However, this methodology, originally described by Rheinwald and Green et al,¹⁷ was found to be quite cumbersome, laborious, and time consuming. We were therefore looking for an easier method of transplantation. This resulted in the development of the micrografting technique.

In 1957, Meek¹⁸ developed a method for micrografting. He devised a system for dividing skin into approximately millimeter-size particles. However, the method was very cumbersome and required placement of the grafts with the dermal-side down. The method therefore never became popular; instead, the expansion of grafts using a mesher developed by Tanner et al¹⁹ became the clinical standard of care. However, whereas Meek¹⁸ could expand skin up to 10 times, the Tanner mesher was not capable of expanding the skin more than approximately 6 times. The theoretical 9-fold expansion makes the skin very difficult to handle.

In porcine experiments, we were able to demonstrate that if you mince skin into submillimeter particles and transplant them into a wet or moist environment, the micrografts will survive regardless of orientation.²⁰ We were also able to show that, in porcine wounds, the micrografts can be expanded 100 times (for instance, a 1-in² graft can cover a 100-in² recipient site) and heal within 14 days (Fig. 7).²⁰ When applied to the entire body, this means theoretically that a piece of skin of approximately 14 × 14 cm could, if divided into micrografts, provide skin grafts for the entire body, for instance, to treat a very large burn.

The micrografting technique also has the advantage of making it easy to transplant wounds that are quite irregular and uneven in shape. The methodology is also helpful for the practitioner and for the facility, in that it can be performed in a wound clinic or doctor's office and by a person without surgical training (Fig. 8).

Kiwanuka et al²¹ compared healing outcomes of full-thickness porcine excisional wounds treated with (1) cultured autologous keratinocytes, (2) split-thickness skin grafts, or (3) micrografts.

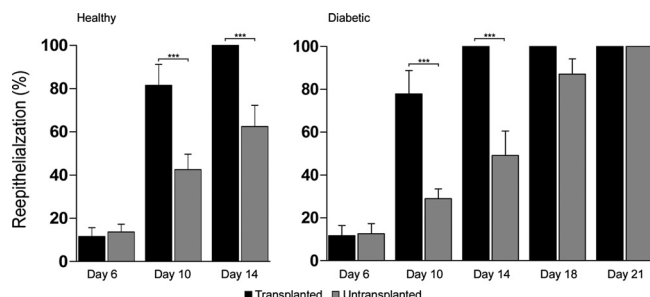


FIGURE 7. Re-epithelialization of porcine wounds. Full-thickness wounds in healthy pigs (left) display accelerated healing after transplantation with minced skin micrografts, with a significantly higher epithelialization rate on day 10 (81.6%) compared with control wounds (42.6%). The difference remains significant on day 14, when all transplanted wounds were re-epithelialized, compared with only 62.5% of the nontransplanted controls. The same results were seen in full-thickness wounds in diabetic pigs (right). (Reprinted with permission from Lippincott Williams & Wilkins.)

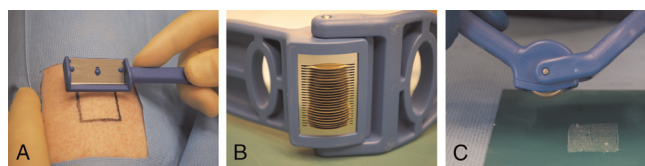


FIGURE 8. Overview of skin micrograft production. A, Single-use dermatome for harvest of skin graft. B, Skin mincer with parallel rotating cutting discs. C, Mincer is run twice over skin graft, yielding micrografts of defined shape and size.

Transplantation of micrografts yielded results similar to split-thickness skin grafts in terms of re-epithelialization rate, scar formation, epidermal morphology, and macroscopic appearance. This was achieved with a skin graft minced into micrografts with immediate 10-fold expansion.

In 2009, during Operation Iraqi Freedom, US Army surgeon Dr. Roy R. Danks²² was presented with an overwhelming burn injury of an Iraqi citizen and did not have the facilities required to treat such an extensive injury. Dr. Danks contacted Dr. Elof Eriksson after he had read an abstract presented at the American Burn Association meeting that described a new method of expanding skin grafts to cover large burns using a simple custom-made skin mincing device. Skin mincers were immediately sent from Brigham and Women's Hospital to Iraq, and a phone consultation followed on their use. The result of this consultation is that this Iraqi civilian has the distinction of surviving one of the highest percentage body surface area burns to be treated outside a nonspecialized burn center.²² In addition, Dr. Svensjö in Malmö, Sweden, has successfully treated both acute and chronic lower-leg ulcers using the micrografting technique (personal correspondence).

Modifying the Microenvironment to Facilitate Wound Healing

The microenvironment of the wound appears to have a significant impact on the speed and quality of healing. Breuing et al⁷ found that porcine partial-thickness wounds healed faster and with fewer aberrations in a wet environment. Vogt et al expanded this study and found that partial-thickness wounds treated in a wet incubator-like environment healed significantly faster than did dry wounds.⁹ Svensjö et al¹⁰ studied full-thickness wounds and found that wet wounds healed significantly faster than did moist wounds, which in turn healed significantly faster than did dry wounds. The studies by Reish et al⁸ revealed a correlation between early local inflammation and the size of the scar at 28 days. It can be theorized that inflammation is reduced through protection from trauma in the wound chamber, avoidance of desiccation, and keeping non-contaminated wound fluid over the wound. The micrometabolism of the wound has not yet been studied, but it appears likely that the caloric losses and the decreased temperature from evaporation of water from the wound surface could have a negative impact on healing.

There is no uniformly accepted safe therapy to reduce scar formation. Containment of the wound during healing creates a platform for the study of potential therapeutic agents and very precise examination of dose-response ratio of administered substances.

The authors have performed experiments with growth media added to the wound and found a trend toward faster healing. However, this study was done in small porcine wounds, which normally heal very quickly. The addition of growth medium to the wound chamber creates a real transition from in vitro to in vivo tissue culture conditions. It also creates a regenerative platform for in vivo

use of tissue-engineering principles. One advantage of this environment is that the blood vessels develop in parallel with the regenerative process. If, in addition, autologous cells are used for the engineering of tissue, there is no immune response; this eliminates the 2 main obstacles encountered by most tissue-engineering strategies.

SUMMARY

This review summarizes experiments performed by us and others, examining the importance of the microenvironment to wound healing. Several studies have compared wet, moist, and dry healing. The authors conclude that a wet, incubator-like microenvironment provides the fastest healing with fewest aberrations and least scar formation. The methodology also creates a platform for the study of agents that potentially increase the speed of healing or reduce scars. With the addition of various active molecules and cells, it also provides a platform technology for regeneration in vivo.

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